Name: _____

TOPIC TEST 1

Chapter 2: Sets and basic probability

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 7 short answer questions (31 marks)
- Part C: 2 analysis questions (14 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 $A = \{n: n = 3m + 1, \text{where } m \in \mathbf{Z}^+\}$, $B = \{n \text{ odd}: 0 \leq n \leq 32\}$.

$A \cap B =$

A {0, 1, 2, …, 29, 30, 31, 32, 34, 37, 40, 43, …}

B {0, 1, 2, …, 29, 30, 31, 32}

C {7, 11, 19, 25, 31}

D {4, 7, 10, 13, 16, 19, 22, 25, 28, 31}

E {7, 13, 19, 25, 31}

[1 mark]

2 A card is drawn from each of two normal packs of playing cards. Which sample space below represents the suits of the cards drawn? (Note that H = hearts, D = diamonds, C = clubs and S = spades.)

A {H, D, C, S, H, D, C, S}

B {HH, HD, HC, HS, DH, DD, DC, DS, CH, CD, CC, CS, SH, SD, SC, SS}

C {HD, HC, HS, DH, DC, DS, CH, CD, CS, SH, SD, SC}

D {HDCS, HDSC, HCDS, HCSD, DHCS, DHSC, DCHS, DCSH, CHDS, CHSD, CDHS, CDSH}

E {HDCS, HDSC, HCDS, HCSD, HSDC, HSCD, DHCS, DHSC, DCHS, DCSH, DSHC, DSCH, CHDS, CHSD, CDHS, CDSH, CSHD, CSDH, SHDC, SHCD, SDHC, SDCH, SCHD, SCDH}

[1 mark]

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3 For the sample space for throwing two eight-sided dice numbered 1–8, $n(\text{total} = 12) =$

A 1

B 8

C 6

D 5

E 4

[1 mark]

4 The probability that a card drawn from a normal pack is a club or an Ace is

A $\frac{4}{13}$

B $\frac{17}{52}$

C $\frac{4}{13}$

D $\frac{1}{17}$

E $\frac{1}{3}$

[1 mark]

5 In a class of 25 students, there are 4 who have no siblings, a set of twins who have another sibling, and 8 students who have one sibling. 7 more students have 2 siblings and the rest have more than 2 siblings. Let $A =$ 'more than 2 siblings'. $P(\overline{A}) =$

A $\frac{6}{25}$

B $\frac{4}{25}$

C $\frac{19}{25}$

D $\frac{21}{25}$

E $\frac{1}{5}$

[1 mark]

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Part B

7 short answer questions
31 marks
Show your working where appropriate.

1 Draw a Venn diagram to show the sets:

a $(A \cap B) \cup \overline{C}$

b $A \cap (B \cup \overline{C})$

[4 marks]

2 Write the following sample spaces.

a The sample space for selecting an Australian state or mainland territory.

b The sample space for choosing two different basic operations (from $+$, $-$, $\times$ and $\div$), with $+-$ and $-+$ being counted as different.

[5 marks]

3 What are the sample space and probabilities for finding the difference of two numbers, each chosen at random from 3, 4, 6 and 7?

[6 marks]

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4 A decorations box contains coloured glass balls on strings. There are 4 green ones, 5 red ones and 2 blue ones. A coloured ball is taken out at random. Work out the probability for each colour.

[3 marks]

5 A box contains mixed nuts of comparable sizes, including 6 mm metric, $\frac{1}{4}''$ Whitworth, 6 mm fine metric and $\frac{1}{4}''$ LH gas thread. To two decimal places, the probabilities of picking Whitworth, 6 mm metric and $\frac{1}{4}''$ LH gas thread nuts are 0.65, 0.25 and 0.03 respectively. The box actually contains 128 nuts. How many are fine metric?

[5 marks]

6 From 90 washing machines checked by the quality control section in a factory, 7 were found to have faults. Estimate the probability of faulty machines and the number you would expect in a month's production of 2500 machines.

[3 marks]

7 In a particular school, 40 Year 12 students are preparing folios to enter one of two local art colleges. 70% are applying to the first college and 10 are applying to both colleges. What is the probability that a student chosen at random is applying to the second college?

[5 marks]

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Part C

2 analysis questions
14 marks
Show your working where appropriate.

1 A rectangle that measures 50 cm by 60 cm is divided 4 smaller rectangles as shown in the diagram. Darts are thrown into the large rectangle to get one of the scores shown.

	40cm	
20 cm	40	20
30 cm	30	5

What are the sample space and probabilities for the scores?

[7 marks]

2 Prove that for mutually exclusive events A and B, $n(A \text{ and } (B \text{ or } C)) = n(A \text{ and } C)$.

[7 marks]
Total marks: 50

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Answers

Part A **[1 mark each]**

1 $A = \{4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, \ldots\}$ and
$B = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31\}$
so $A \cap B = \{7, 13, 19, 25, 31\}$.
$\therefore$ **E** **[1 mark]**

2 For H first, the second can be H, D, C, S, and the same for each other suit, so
$S = \{HH, HD, HC, HS, DH, DD, DC, DS, CH, CD, CC, CS, SH, SD, SC, SS\}$.
$\therefore$ **B** **[1 mark]**

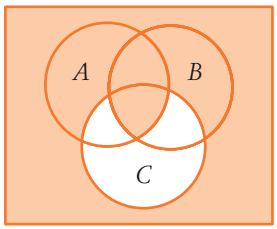
3 The event, 'total = 12' = {48, 57, 66, 75, 84}, so $n(\text{total} = 12) = 5$.
$\therefore$ **D** **[1 mark]**

4 $n(\text{club}) = 13$, $n(\text{ace}) = 4$ and $n(\text{ace of clubs}) = 1$, so $n(\text{club or ace}) = 13 + 4 - 1 = 16$.
Thus $P(\text{club or ace}) = \frac{16}{52} = \frac{4}{13}$.
$\therefore$ **A** **[1 mark]**

5 $n(0 \text{ siblings}) = 4$, $n(1 \text{ sibling}) = 8 + 2 = 10$, $n(2 \text{ siblings}) = 7$, so $n(\text{more than 2 siblings})$
$= 25 - (10 + 4 + 7) = 4$. Thus $P(\overline{A}) = \frac{21}{25}$.
$\therefore$ **D** **[1 mark]**

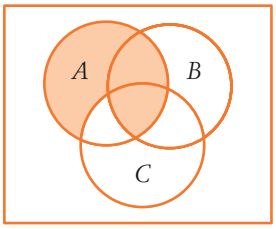
Part B

1 a



[2 marks]

b



[2 marks]

[4 marks]

2 a {Vic., NSW, Qld, WA, Tas., NT, SA, ACT} **[2 marks]**

b $\{+\times, +-, +\div, -+, -\times, -\div, \times+, \times-, \times\div, \div+, \div-, \div\times\}$ **[3 marks]**

[5 marks]

3 $S = \{3-3, 3-4, 3-6, 3-7, 4-3, 4-4, 4-6, 4-7, 6-3, 6-4, 6-6, 6-7, 7-3, 7-4, 7-6, 7-7\}$ **[2 marks]**

$P(-4) = \frac{1}{16}$, $P(-3) = \frac{1}{8}$, $P(-2) = \frac{1}{16}$, $P(-1) = \frac{1}{8}$, $P(0) = \frac{1}{4}$, $P(1) = \frac{1}{8}$, $P(2) = \frac{1}{16}$,
$P(3) = \frac{1}{8}$, $P(4) = \frac{1}{16}$, **[4 marks]**

[6 marks]

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4 $P(\text{green}) = \frac{4}{11}$, $P(\text{red}) = \frac{5}{11}$, $P(\text{blue}) = \frac{2}{11}$ **[1 mark each]**

[3 marks]

5 $P(\text{fine metric}) = 1 - (0.65 + 0.25 + 0.03)$ **[1 mark]**

$= 0.07$ **[1 mark]**

$P(\text{fine metric}) = \frac{n(\text{fine metric})}{n(S)}$, so $0.07 = \frac{n(\text{fine metric})}{128}$ **[1 mark]**

so $n(\text{fine metric}) = 0.07 \times 128 \approx 9$ **[1 mark]**

There are 9 fine metric nuts. **[1 mark]**

[5 marks]

6 $P(\text{faulty}) = \frac{7}{90}$ **[1 mark]**

Faulty from 2500 $= \frac{7}{90} \times 2500 \approx 194$ **[1 mark]**

About 194 faulty machines will be expected. **[1 mark]**

[3 marks]

7 EITHER:

$P(\text{both colleges}) = \frac{10}{40} = 0.25$ **[1 mark]**

$P(\text{either college}) = P(\text{first college}) + P(\text{second college}) - P(\text{both colleges})$ **[1 mark]**

$1 = 0.7 + P(\text{second college}) - 0.25$ **[1 mark]**

so $P(\text{second college}) = 0.55$ **[1 mark]**

The probability that a student is applying for the second college is 0.55. **[1 mark]**

[5 marks]

OR

Number applying for first college = 70% of 40 = 28 **[1 mark]**

Number applying for first college but not second = 28 − 10 = 18 **[1 mark]**

Number applying for second college = 40 − 18 = 22 **[1 mark]**

Probability of applying for second college $= \frac{22}{40} = 0.55$ **[2 marks]**

[5 marks]

OR

Steps of the second method on a Venn diagram.

Part C

1 Assume that the probabilities are proportional to the areas and the darts land in the rectangle. **[1 mark]**

Area (40) $= 20 \times 20 = 400$ cm^2

Area (30) $= 20 \times 30 = 600$ cm^2

Area (20) $= 20 \times 40 = 800$ cm^2

Area (5) $= 30 \times 40 = 1200$ cm^2 **[1 mark]**

Total area $= 50 \times 60 = 3000$ cm^2

Then $S = \{5, 20, 30, 40\}$ **[1 mark]**

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$P(5) = \frac{1200}{3000} = \frac{2}{5}$, $P(20) = \frac{800}{3000} = \frac{4}{15}$, $P(30) = \frac{600}{3000} = \frac{1}{5}$, $P(40) = \frac{400}{3000} = \frac{2}{15}$ **[1 mark each]**

[7 marks]

2 Since A and B are mutually exclusive, $n(A \text{ and } B) = 0$.

From $P(A \text{ and } B) = P(A) + P(B) - P(A \text{ or } B)$, $n(A \text{ and } B) = n(A) + n(B) - n(A \text{ or } B)$,

so $n(A \text{ and } (B \text{ or } C)) = n(A) + n(B \text{ or } C) - n(A \text{ or } (B \text{ or } C))$ **[1 mark]**

$= n(A) + n(B \text{ or } C) - n(A \text{ or } B \text{ or } C)$

$= n(A) + n(B \text{ or } C) - n((A \text{ or } B) \text{ or } C)$ **[1 mark]**

Using $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B)$,

$n(A \text{ and } (B \text{ or } C)) = n(A) + n(B \text{ or } C) - n((A \text{ or } B) \text{ or } C)$

$= n(A) + n(B) + n(C) - n(B \text{ and } C) - (n(A \text{ or } B) + n(C) - n((A \text{ or } B) \text{ and } C))$

[1 mark]

Because A and B are mutually exclusive,

$n((A \text{ or } B) \text{ and } C) = n(A \text{ and } C) + n(B \text{ and } C)$ and $n(A \text{ or } B) = n(A) + n(B)$ **[1 mark]**

So $n(A \text{ and } (B \text{ or } C)) = n(A) + n(B) + n(C) - n(B \text{ and } C)$

$- (n(A) + n(B) + n(C) - n(A \text{ and } C) - n(B \text{ and } C))$ **[1 mark]**

$= n(A) + n(B) + n(C) - n(B \text{ and } C) - n(A) - n(B) - n(C)$

$+ n(A \text{ and } C) + n(B \text{ and } C)$ **[1 mark]**

$= n(A \text{ and } C)$ **QED** **[1 mark]**

[7 marks]

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